

19 Turbot in Skagerrak and Kattegat

tur.27.3a – *Scophthalmus maximus* in Division 3.a

Turbot (*Scophthalmus maximus*) is assessed in the Working Group on the Assessment of Demersal Stocks in the North Sea and Skagerrak (WGNSSK). The advice issued in 2017 for the years 2018 and 2019 was based on the “2/3 rule” for category 3 stocks, applied to the IBTS Q1 and Q3 biomass indices. In 2019 and 2020, ICES was not requested to provide advice on fishing opportunities for this stock and the advice sheet reported only on the status of the stock. From 2021 onwards, ICES is requested to provide advice again and on an annual basis.

The general perception is that landings have fluctuated without clear trend. In 2017, length-based indicators (LBI) and exploratory SPiCT runs were examined, pointing out that the stock may be exploited sustainably. In 2019, the survey indices were of poor quality, with low catch rates and large annual fluctuations, and they showed no clear trends. The LBI indicators were not updated due to poorer length information available following reduced sampling since 2017. The stock was benchmarked in 2020 where a SPiCT assessment was accepted to provide information on stock status ([ICES, 2020a](#)). In 2025, the SPiCT assessment was revised as the benchmark model showed poor performance (WD 8 in Annex 8). The revised assessment is used since 2025 to provide catch advice according to the MSY approach (Category 2 stock).

19.1 Management regulations

Turbot in Division 3.a. is not managed using a TAC. The catch advice for 2026 is 395 tonnes.

There is no official EU minimum landing size (MLS). In the Netherlands, various restrictions and MLS for North Sea turbot have been applied by Dutch POs over time, which may also affect the Dutch discarding of turbot caught in Skagerrak.

19.2 Fisheries data

Turbot is now only caught as bycatch in the trawl and gillnet fisheries. Table 19.1 and Figure 19.1 summarize turbot landings in ICES Division 3.a. Since 1950, total landings (3.a) ranged from 64 to 736 tonnes. The lowest landings were recorded in the 1960s and the highest peaks are observed in the late 1970s and in the early 1990s. The peak in the 1970s is linked to exceptionally high records from the Netherlands for four years. No official landings are reported from the Netherlands prior to 1975. The Danish landings, present throughout the time series, have fluctuated without trend with an average around 150 tonnes per year. In the last decades, the total annual landings of turbot in 3.a declined from 300–400 tonnes in the early 1990s to around 100 tonnes in the early 2010s but have increased again in the most recent years.

The total landings imported in InterCatch for 2024 were 147 tonnes. The procedure to raise discards for métiers that do not have them reported is done by grouping fleets into three groups. For each group the mean discard rate is calculated from métiers in that group that have reported discards, and that weighted mean is used to raise discards for the rest of the métiers. Métiers that have discard rates above 2 are not included in the calculation of the mean. The three groups are: all fleets in subdivision 20, all fleets in subdivision 21, métiers from the Netherlands in Division 3.a. The Norwegian fleet and the industrial fleet are assumed to have no discards. There is a total

of 6 tonnes of imported and raised discards in 2024. The total catch in 2024 was 153 tonnes (Table 19.2; Table 19.3).

The stock was benchmarked in 2020, which included a data call for turbot in Division 3.a that led to new landings and discard data being uploaded to InterCatch. This allowed a compilation of information by area and metier. During the benchmark, reported discard ratios were available across the period 2002–2018, and the average discard ratio (10.49%) was used to reconstruct the discards for earlier years (1950–2001). Details of the benchmark are provided in the associated report ([ICES, 2020a](#)).

Discard coverage was on average 67% in 3a.20 and 49% in 3a.21 between 2002–2019. In 2020, coverage was lower in Subdivision 3a.20 (47%) and higher in 3a.21 (59%). The discard coverage in 2021 and 2022 was higher in both subdivisions. The coverage was at similar levels in 2024, with 55% for Subdivision 3a.20 and 73% for Subdivision 3a.21. The beam trawl fleet from the Netherlands and the gillnet fleet from Denmark are the largest metiers without discard information (Figure 19.2). Discarding is clearly related to fish size, most individuals below 30 cm are being discarded (Figure 19.3).

As turbot in 3a is mainly a bycatch species, a change in catch over time can be influenced by changes in effort levels and targeting of the fleets in the area that catch it. Further investigation is needed into targeting of the species in the area through time.

19.3 Survey data, recruit series and analysis of stock trends

A new exploitable biomass index was developed during the 2025 revision of the assessment model (WD 8 in Annex 8). The updated model includes an explicit Year $\times$ Quarter interaction, which allows the model to capture interannual and seasonal variability directly. This model is used to predict three survey index time-series, one each for Q1, Q3, and Q4. These time-series are used as input to the SPiCT model.

The updated model is derived from a compilation of five surveys conducted in Division 3a. More specifically, the surveys included the beam trawl survey (BTS), the North Sea International Bottom Trawl Survey (NS-IBTS), the Baltic International Trawl Survey (BITS), a Danish national survey targeting sole and a Swedish-Danish survey targeting cod, all covering parts of Division 3.a. Since the index was intended for use in SPiCT, only the vulnerable sizes of the individuals caught in the surveys were included in the calculation of the index, leading to an exploitable biomass index. Length distributions from the commercial fleet and scientific surveys were used to derive an exploitable biomass index. This was done using the ratio of total commercial catch at length to survey catch at length for the period 2012–2018 to down-weight the smaller length groups in the survey, such that the survey can be considered representative for the exploitable stock biomass and thus suitable for use in a biomass production model. The standardised exploitable biomass indices are shown in Figure 19.4, along with 5 retrospective runs, calculated by leaving out the last 1–5 years of available data. The SPiCT model combined the three exploitable biomass indices and updated fisheries data and was approved after external review by WGNSSK in 2025 (WD 8 in Annex 8).

19.4 Assessment and short-term forecast

The surplus production model in continuous time (SPiCT, Pedersen and Berg 2017) is used for the assessment of the stock. The main settings are:

Setting/Data	Values/Source
Catch time-series	1960–2024
Combined survey index Q1	1983–2024
Combined survey index Q3	1991–2024
Combined survey index Q4	1991–2024
SPiCT settings	
Standard deviation on the catch (observation) (stdevfacC)	From 1960–2001 stdevfacC = 3; From 2001–present stdevfacC = 1
Standard deviation on the indices (observation) (sdi in Q1, Q3 and Q4)	$\log(sdi) \sim N(\log(lsd), 0.5^2)$, where l_{sd} represents the standard deviation of the index time series for the respective quarter
Uncertainty ratio of index (observation) to biomass process (alpha)	Deactivated
Uncertainty ratio of catch (observation) to fishing mortality process (beta)	Deactivated
Shape parameter (n)	n is assigned a gamma prior with a mode of 1.478, where the probability of n being less than 1 is zero
Intrinsic growth rate (r)	$\log(r) \sim N(\log(0.5), 0.5^2)$
Initial depletion (bkfrac)	$\log(bkfrac) \sim N(\log(0.75), 1^2)$
Discretion time-step (dteuler)	1/16 year (default)

The assessment results are shown in Figure 19.5 and summarised in Table 19.5. The relative biomass fluctuates without trend and is estimated above the reference point. Relative fishing mortality is estimated below the reference point. It has decreased since 2000 and is stable at a lower level in most recent years. The diagnostics of the goodness of fit of the model are based on the one-step-ahead residuals (Figure 19.6). The prior-posterior distribution of the priors is presented in Figure 19.7. Sensitivity to initial parameters, using the *check.ini* function, shows that all converged runs return the same parameter estimate.

The retrospective analysis is done using the recalculated index by consecutively removing the last five years of survey information and then fitting the assessment model. Both the relative fishing biomass (B/B_{MSY}) and relative fishing mortality (F/F_{MSY}) estimates have acceptable retrospective bias (Mohn's rho: -0.015 for B/B_{MSY} and 0.059 for F/F_{MSY} ; Figure 19.8).

A short-term forecast is performed using SPiCT. The assumption for the intermediate year (2025) is that the fishing mortality process continues, essentially keeping status quo fishing mortality. This leads to the following intermediate year assumptions:

Variable	Value	Notes
F_{2025}/F_{MSY}	0.35	Status quo F: F_{2024}/F_{MSY}
B_{2026}/B_{MSY}	1.59	Short-term forecast (STF)
Catch (2025)	167	STF of catch under F_{sq} ; in tonnes
Discard rate (2025, 2026)	8.9	Average discard rate by weight 2022–2024; in %

Variable	Value	Notes
Projected landings (2025)	152	STF; assuming average landings ratio; in tonnes
Projected discards (2025)	15	STF; assuming average discards ratio; in tonnes

To provide advice following the MSY approach, the recommendation of WKLIFE X (ICES, 2020b) is followed. The basis for the advice assumes fishing mortality $F=F_{MSY}$ during the management year (2026). Then the catch advice is the 35th percentile of the projected catch distribution. The use of that percentile instead of the median leads to a more precautionary advice, with no loss of long-term yield and incorporates the uncertainty into the advice. For 2026, the catch advice is 395 tonnes. The results for the baseline management scenario and alternatives are shown in Figure 19.9 and Table 19.4.

The change in advice compared with last year's advice (261 tonnes) is large (+51%). The change is due to a revised assessment model in 2025 (WD 8 in Annex 8), which indicates an improved stock status characterized by a lower F/F_{MSY} and a higher B/B_{MSY} . The difference between last year's assessment and this year's assessment are relatively large due to a revised assessment model, which estimates a lower F/F_{MSY} and a higher B/B_{MSY} (Figure 19.10).

19.5 Issue list

The stock was revised in 2025, but some issues remain:

- Stock identity. The benchmark indicated that Division 3.a is not a separate stock, but connected to both the North Sea and the Baltic Sea. There is genetic differentiation between the North Sea and the Baltic Sea with a genetic hybrid zone within Division 3.a. The new exploitable biomass index and the landings data indicated elevated abundances and landings on the borders between Division 3.a and the North Sea and the Baltic Sea, further supporting connectivity between Division 3.a and neighbouring areas. The stock identity of Division 3.a is currently evaluated with genetic analysis (in preparation).
- The amount of length distribution data significantly reduced since 2017. Denmark is responsible for approximately 3/4 of the turbot landings in Division 3.a. Length information collected by Denmark is highly valuable for this stock and should be considered when drafting the next data collection programme. The validity of using length distributions from the commercial fleet and scientific surveys from 2012-2018 to derive an exploitable biomass index should be tested using more recent data.
- Time series of catch and survey data. The peak in Dutch turbot landings in Division 3.a in late 1970s could warrant further investigation. However, note that excluding catch data prior to 1980 does not affect SPiCT's estimates of the recent stock status. In addition, Cardinale *et al.* (2009) reconstructed a long time series of survey data. It would be interesting to investigate options to include it in SPiCT runs. The paper indicated historic declines in abundance and maximum body sizes of turbot in Division 3.a.

19.6 Summary

The assessment of turbot stock in Division 3.a was revised in 2025, and the resulting SPiCT model was used for the present assessment and report. A major improvement for the SPiCT model was the development of a model with three indices for the relative exploitable biomass based on five surveys covering Division 3.a. The analyses indicate that the relative exploitable biomass (B/B_{MSY})

is above the reference point of 0.5 and relative fishing mortality (F/F_{MSY}) below the reference point of 1.

19.7 References

- Cardinale, M., Svedäng, H., Bartolino, V., Maiorano, L., Casini, M., Hjelm, J. Linderholm H. 2012. Spatial and temporal depletion of haddock and pollack during the last century in the Kattegat-Skagerrak. *Journal of Applied Ichthyology*, 1–12. <https://doi.org/10.1111/j.1439-0426.2012.01937.x>
- ICES. 2020a. Benchmark Workshop for Flatfish stocks in the North Sea and Celtic Sea (WKFlatNSCS). ICES Scientific Reports. 2:23. 975 pp. <http://doi.org/10.17895/ices.pub.5976>
- ICES. 2020b. Tenth Workshop on the Development of Quantitative Assessment Methodologies based on LIFE-history traits, exploitation characteristics, and other relevant parameters for data-limited stocks (WKLIFE X). ICES Scientific Reports. 2:98. 72 pp. <http://doi.org/10.17895/ices.pub.5985>
- Pedersen, M. W., and Berg, C. W. 2017. A stochastic surplus production model in continuous time. *Fish and Fisheries*, 18: 226–243. <https://doi.org/10.1111/faf.12174>

19.8 Tables and figures

Table 19.1. Turbot in Division 3.a. History of commercial landings; the official landings and below minimum size (BMS) landings are presented for each country participating in the fishery. All weights are in tonnes.

Year	Bel- gium	Ger- many	Den- mark	UK	Nether- lands	Nor- way	Swe- den	BMS	Total
1950	0	13	212	0	0	1	73		299
1951	0	6	191	0	0	6	62		265
1952	0	6	114	0	0	3	58		181
1953	0	4	80	0	0	4	51		139
1954	0	0	78	0	0	1	61		140
1955	0	4	77	0	0	0	49		130
1956	0	7	75	0	0	0	41		123
1957	0	3	108	0	0	0	30		141
1958	0	7	112	0	0	0	41		160
1959	0	6	132	0	0	3	43		184
1960	0	11	115	0	0	2	46		174
1961	0	4	130	0	0	0	45		179
1962	0	5	157	0	0	0	0		162
1963	0	4	124	0	0	0	0		128
1964	0	5	89	0	0	0	0		94
1965	0	6	79	1	0	0	0		86
1966	0	2	104	0	0	0	0		106
1967	0	4	68	1	0	0	0		73
1968	0	0	64	0	0	0	0		64
1969	0	1	75	0	0	0	0		76
1970	0	1	76	0	0	0	0		77
1971	0	1	100	0	0	0	0		101
1972	0	2	130	0	0	0	0		132
1973	0	2	98	0	0	0	0		100
1974	0	1	116	0	0	0	0		117
1975	0	2	167	0	7	0	7		183

Year	Bel- gium	Ger- many	Den- mark	UK	Nether- lands	Nor- way	Swe- den	BMS	Total
1976	7	2	178	0	190	0	6		383
1977	7	4	331	0	389	0	5		736
1978	2	4	327	0	186	0	6		525
1979	8	0	307	0	87	0	4		406
1980	7	0	205	1	14	0	6		233
1981	2	0	183	2	12	0	8		207
1982	1	0	164	1	9	0	7		182
1983	4	0	171	0	24	0	10		209
1984	0	0	176	0	0	0	12		188
1985	1	0	224	0	0	0	16		241
1986	2	0	180	0	0	0	11		193
1987	5	0	147	0	0	0	9		161
1988	2	0	115	0	11	0	10		138
1989	2	0	173	0	0	0	9		184
1990	5	0	363	0	0	0	18		386
1991	4	0	244	0	0	7	21		276
1992	4	0	278	0	0	8	19		309
1993	3	2	336	0	0	10	0		351
1994	2	1	313	0	0	15	22		353
1995	4	1	268	0	0	17	11		301
1996	0	1	185	0	0	13	11		210
1997	0	0	200	0	0	9	11		220
1998	0	1	148	0	0	7	8		164
1999	0	1	139	0	0	10	6		156
2000	0	1	180	0	0	6	6		193
2001	0	0	227	0	0	8	3		238
2002	0	1	205	0	0	11	5		222
2003	0	0	128	0	13	14	4		159
2004	0	0	119	0	14	7	7		147

Year	Bel- gium	Ger- many	Den- mark	UK	Nether- lands	Nor- way	Swe- den	BMS	Total
2005	0	0	108	0	7	6	6		127
2006	0	1	95	0	8	8	9		121
2007	0	1	138	0	15	7	12		173
2008	0	1	121	0	4	6	11		143
2009	0	1	94	0	2	6	17		120
2010	0	0	72	0	6	4	13		95
2011	0	1	78	0	0	7	13		99
2012	0	< 0.5	167	0	0	8	14		189
2013	0	< 0.5	91	0	0	5	15		111
2014	0	1	94	0	3	6	18		122
2015	0	< 0.5	135	0	20	8	11		174
2016	0	< 0.5	137	< 0.5	25	6	11		179
2017	0	< 0.5	154	0	16	7	12	< 0.5	189
2018	0	0	109	0	24	8	10	< 0.5	151
2019	0	0	117	<0.5	69	5	7	< 0.5	198
2020	0	0	124	0	55	5	7	< 0.5	191
2021	0	1	139	0	39	5	8	0	192
2022	0	0	106	0	20	4	6	< 0.5	137
2023*	0	1	113	0	11	6	7	< 0.5	139
2024*	0	< 0.5	101	0	32	5	9	< 0.5	147

* Preliminary

Table 19.2. Turbot in Division 3.a. Landings and discards (in kg) by year and area after discard raising in InterCatch (using CATON estimate). No BMS nor logbook registered discards reported in InterCatch.

Year	Discards	Landings	Total	discard ratio
2002	17593	214745	232338	7.60%
27.3.a	9	135	144	6.20%
27.3.a.20	906	152506	153412	0.59%
27.3.a.21	16679	62104	78783	21%
2003	15273	153228	168501	9.10%
27.3.a	1468	14080	15548	9.40%

Year	Discards	Landings	Total	discard ratio
27.3.a.20	227	83702	83929	0.27%
27.3.a.21	13578	55446	69024	19.70%
2004	9463	146736	156199	6.10%
27.3.a	990	15674	16664	5.90%
27.3.a.20	2524	72802	75326	3.40%
27.3.a.21	5950	58260	64210	9.30%
2005	10672	125757	136429	7.80%
27.3.a	516	6928	7444	6.90%
27.3.a.20	3277	73824	77101	4.30%
27.3.a.21	6880	45005	51885	13.30%
2006	11600	116895	128495	9.00%
27.3.a	833	8838	9671	8.60%
27.3.a.20	246	55105	55351	0.44%
27.3.a.21	10522	52952	63474	16.60%
2007	32300	171442	203742	15.90%
27.3.a	1597	16098	17695	9.00%
27.3.a.20	880	100442	101322	0.87%
27.3.a.21	29823	54902	84725	35%
2008	7183	139685	146868	4.90%
27.3.a	172	4635	4807	3.60%
27.3.a.20	0	91024	91024	0.00%
27.3.a.21	7011	44026	51037	13.70%
2009	9363	120692	130055	7.20%
27.3.a	142	2661	2803	5.10%
27.3.a.20	727	73619	74346	0.98%
27.3.a.21	8494	44412	52906	16.10%
2010	11264	96525	107789	10.50%
27.3.a	658	6346	7004	9.40%
27.3.a.20	163	43069	43232	0.38%

Year	Discards	Landings	Total	discard ratio
27.3.a.21	10443	47110	57553	18.10%
2011	25532	94354	119886	21%
27.3.a	59	258	317	18.60%
27.3.a.20	4192	54053	58245	7.20%
27.3.a.21	21281	40042	61323	35%
2012	22621	194736	217357	10.40%
27.3.a	29	289	318	9.10%
27.3.a.20	3562	164297	167859	2.10%
27.3.a.21	19030	30150	49180	39%
2013	7110	110945	118055	6.00%
27.3.a	0	2	2	0.00%
27.3.a.20	1469	75803	77272	1.90%
27.3.a.21	5641	35140	40781	13.80%
2014	14520	122406	136926	10.60%
27.3.a	0	0	0	0.00%
27.3.a.20	3874	82446	86320	4.50%
27.3.a.21	10646	39960	50606	21%
2015	33938	179737	213675	15.90%
27.3.a	0	1	1	0.00%
27.3.a.20	8426	141894	150320	5.60%
27.3.a.21	25511	37842	63353	40%
2016	19246	190829	210075	9.20%
27.3.a	3492	34530	38022	9.20%
27.3.a.20	9617	111770	121387	7.90%
27.3.a.21	6136	44529	50665	12.10%
2017	31669	191667	223336	14.20%
27.3.a	2928	17528	20456	14.30%
27.3.a.20	17404	122493	139897	12.40%
27.3.a.21	11337	51646	62983	18.00%

Year	Discards	Landings	Total	discard ratio
2018	22528	153398	175926	12.80%
27.3.a	4000	24842	28842	13.90%
27.3.a.20	11506	82913	94419	12.20%
27.3.a.21	7022	45643	52665	13.30%
2019	41903	204356	246259	17.00%
27.3.a	15857	74430	90287	17.60%
27.3.a.20	21409	102564	123973	17.30%
27.3.a.21	4637	27362	31999	14.50%
2020	13458	201698	215156	6.3%
27.3.a	4673	65140	69813	6.7%
27.3.a.20	3210	106819	110029	2.9%
27.3.a.21	5575	29740	35315	15.8%
2021	7682	195050	202732	3.8%
27.3.a	1663	42090	43753	3.8%
27.3.a.20	1211	117820	119031	1.02%
27.3.a.21	4807	35140	39947	12.0%
2022	36197	142855	179052	20%
27.3.a	7608	23960	31568	24%
27.3.a.20	18317	91505	109822	16.7%
27.3.a.21	10272	27390	37662	27%
2023	3505	140753	144258	2.4%
27.3.a	-	-	-	-
27.3.a.20	2408	117471	119879	2.0%
27.3.a.21	1097	23283	24380	4.5%
2024	5937	147139	153076	3.9%
27.3.a	-	-	-	-
27.3.a.20	2104	118866	120970	1.74%
27.3.a.21	3832	28274	32106	12%

Table 19.3. Turbot in Division 3.a. Summary of the imported/raised data for 2024. Stock exported without length allocation. Weights are in kg.

Discards	5937	
Imported Data	3572	60%
Raised Discards	2365	40%
Landings	147139	
Imported Data	147139	
Grand Total	153076	

Table 19.4. Turbot in Division 3.a. Forecast table for the baseline and alternative scenarios. The percent B/B_{MSY} change refers to the B/B_{MSY} in 2027 relative to 2026.

Basis	Total catch (2026)	Projected landings (2026)	Projected discards (2026)	Fishing mortality F_{2026}/F_{MSY}	Stock size B_{2027}/B_{MSY}	% B/B_{MSY} change
MSY approach (35 th percentile of predicted catch distribution under $F = F_{MSY}$)	395	360	35	0.85	1.45	-8.9
Other scenarios						
F_{MSY}	455	414	41	1.00	1.39	-12.2
$F = F_{2025}$	174	158	16	0.35	1.64	3.2
$F = 0$	0	0	0	0	1.79	12.7
$F = F_{MSY}$, all fractiles	326	297	29	0.69	1.51	-5.1

Table 19.5. Turbot in Division 3.a. Assessment summary. The 2025 biomass is the short-term forecast during the intermediate year, assuming that the F process continues unchanged from the last year with observations (F_{sq}). Since 2018, discards include BMS landings from EU fleets. Discards are reconstructed prior to 2002.

Year	Relative exploitable biomass			Landings tonnes	Discards tonnes	Relative fishing pressure		
	Low	B/B_{MSY}	High			Low	F/F_{MSY}	High
1960	0.69	2.5	8.9	174	21	0.090	0.28	0.87
1961	0.9	2.3	5.9	179	21	0.108	0.29	0.76
1962	1.03	2.2	4.6	162	19	0.107	0.26	0.62
1963	1.11	2.1	3.8	128	15	0.091	0.21	0.46
1964	1.15	1.96	3.3	94	11	0.078	0.171	0.37
1965	1.19	1.92	3.1	86	10	0.083	0.177	0.38
1966	1.27	1.98	3.1	106	13	0.079	0.167	0.35
1967	1.28	1.95	3.0	73	9	0.063	0.133	0.28

Year	Relative exploitable biomass			Landings	Discards	Relative fishing pressure		
	Low	B/B _{MSY}	High			Low	F/F _{MSY}	High
1968	1.25	1.88	2.8	64	8	0.064	0.133	0.28
1969	1.28	1.91	2.9	76	9	0.070	0.145	0.30
1970	1.3	1.93	2.9	77	9	0.079	0.163	0.34
1971	1.32	1.96	2.9	101	12	0.103	0.21	0.43
1972	1.36	2.0	3.0	132	16	0.107	0.22	0.45
1973	1.31	1.94	2.9	100	12	0.103	0.21	0.43
1974	1.26	1.87	2.8	117	14	0.136	0.27	0.55
1975	1.29	1.91	2.8	183	22	0.23	0.46	0.93
1976	1.36	2.0	3.0	383	46	0.44	0.93	1.97
1977	1.30	2.0	3.1	736	88	0.57	1.36	3.3
1978	0.96	1.66	2.9	525	63	0.47	1.26	3.4
1979	0.69	1.33	2.6	406	49	0.34	0.99	2.9
1980	0.55	1.13	2.4	233	28	0.25	0.73	2.2
1981	0.53	1.10	2.2	207	25	0.22	0.63	1.81
1982	0.58	1.13	2.2	182	22	0.22	0.60	1.66
1983	0.67	1.21	2.2	209	25	0.23	0.60	1.61
1984	0.67	1.17	2.1	188	23	0.23	0.60	1.56
1985	0.77	1.31	2.2	241	29	0.23	0.59	1.52
1986	0.8	1.34	2.2	193	23	0.186	0.47	1.20
1987	0.85	1.40	2.3	161	19	0.157	0.39	0.96
1988	0.87	1.39	2.2	138	17	0.156	0.37	0.88
1989	0.98	1.53	2.4	184	22	0.22	0.50	1.16
1990	1.22	1.85	2.8	386	46	0.26	0.60	1.38
1991	1.35	2.1	3.2	276	33	0.23	0.54	1.27
1992	1.28	1.91	2.8	309	37	0.28	0.65	1.51
1993	1.23	1.83	2.7	351	42	0.31	0.72	1.68
1994	1.17	1.75	2.6	353	42	0.30	0.72	1.71
1995	1.11	1.71	2.6	301	36	0.25	0.60	1.46

Year	Relative exploitable biomass			Landings	Discards	Relative fishing pressure		
	Low	B/B _{MSY}	High	tonnes	tonnes	Low	F/F _{MSY}	High
1996	0.98	1.50	2.3	210	25	0.22	0.54	1.31
1997	0.96	1.49	2.3	220	26	0.22	0.53	1.30
1998	0.87	1.34	2.1	164	20	0.198	0.48	1.16
1999	0.78	1.20	1.85	156	19	0.22	0.53	1.27
2000	0.79	1.21	1.85	193	23	0.28	0.68	1.62
2001	0.76	1.17	1.79	238	28	0.31	0.73	1.75
2002	0.71	1.10	1.7	215	18	0.23	0.56	1.33
2003	0.77	1.19	1.84	153	15	0.166	0.39	0.92
2004	0.86	1.31	2.0	147	9	0.134	0.31	0.74
2005	1.05	1.61	2.5	126	11	0.110	0.26	0.59
2006	1.06	1.60	2.4	117	12	0.136	0.31	0.73
2007	1.14	1.72	2.6	171	32	0.154	0.36	0.84
2008	1.10	1.66	2.5	140	7	0.115	0.27	0.62
2009	1.09	1.64	2.5	121	9	0.103	0.24	0.55
2010	1.10	1.64	2.4	97	11	0.088	0.20	0.46
2011	1.13	1.67	2.5	94	26	0.125	0.29	0.65
2012	1.29	1.92	2.9	195	23	0.132	0.30	0.70
2013	1.24	1.86	2.8	111	7	0.088	0.20	0.46
2014	1.20	1.77	2.6	122	15	0.114	0.26	0.59
2015	1.50	2.2	3.3	180	34	0.134	0.31	0.71
2016	1.58	2.4	3.6	191	19	0.146	0.34	0.77
2017	1.45	2.1	3.2	192	32	0.144	0.33	0.75
2018	1.30	1.93	2.9	153	23	0.154	0.35	0.81
2019	1.26	1.87	2.8	204	42	0.183	0.42	0.98
2020	1.25	1.88	2.8	202	13	0.166	0.39	0.90
2021	1.16	1.74	2.6	195	8	0.187	0.43	1.01
2022	0.97	1.45	2.2	143	36	0.167	0.39	0.90
2023	0.9	1.35	2.0	141	4	0.149	0.34	0.80

Year	Relative exploitable biomass			Landings	Discards	Relative fishing pressure		
	Low	B/B _{MSY}	High			Low	F/F _{MSY}	High
2024	0.89	1.34	2.0	147	6	0.148	0.35	0.84
2025	0.99	1.52	2.3					

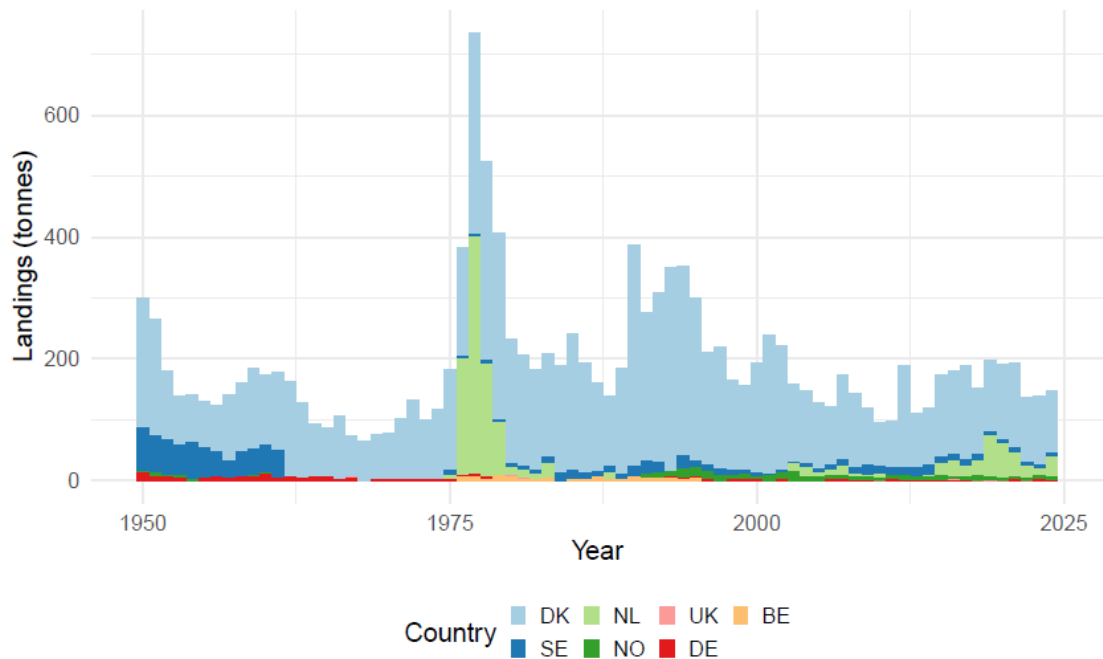


Figure 19.1. Turbot in Division 3.a. Official landings by country since 1950.

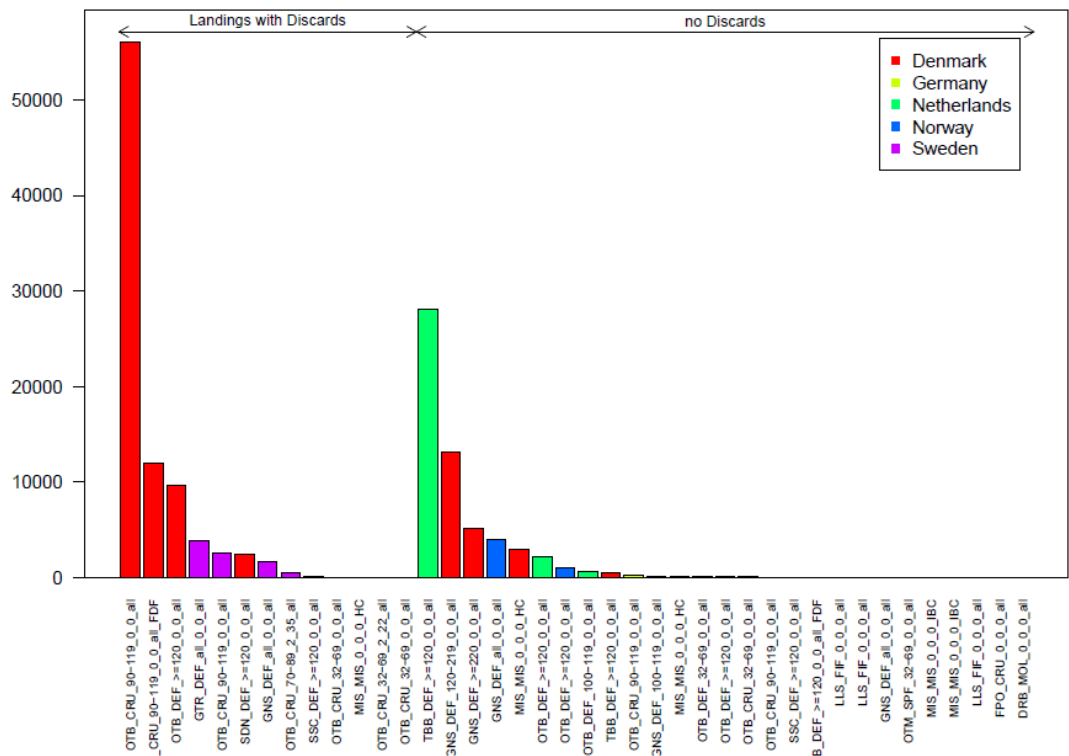


Figure 19.2. Turbot in Division 3.a. Summary of the information provided to InterCatch for 2024. Landings by metier and country, distinguishing between strata with and without corresponding discard information provided.

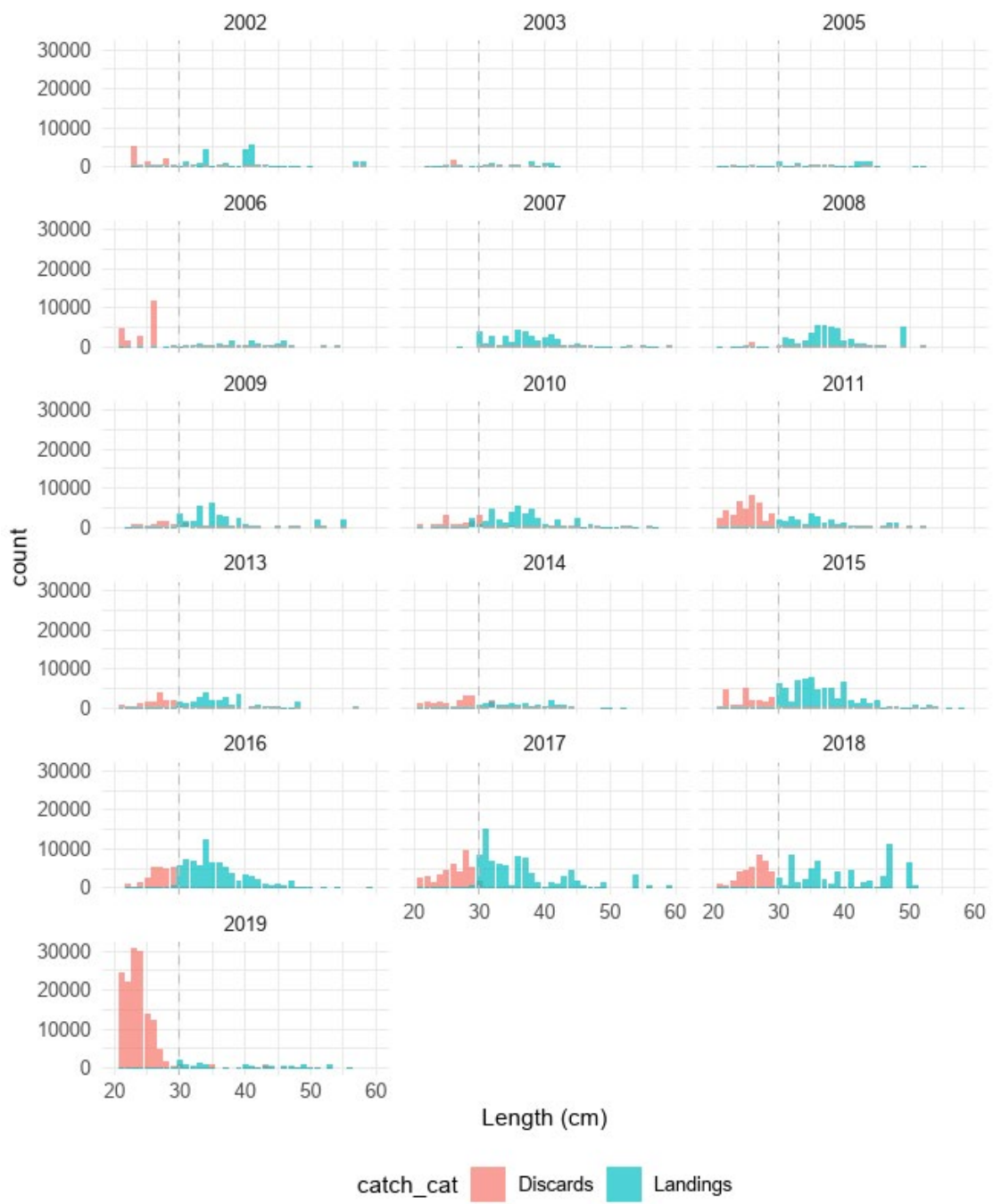


Figure 19.3. Turbot in Division 3.a. Length distribution in landings and discards for the period 2002–2019. Most individuals below 30 cm are discarded (vertical dashed line).

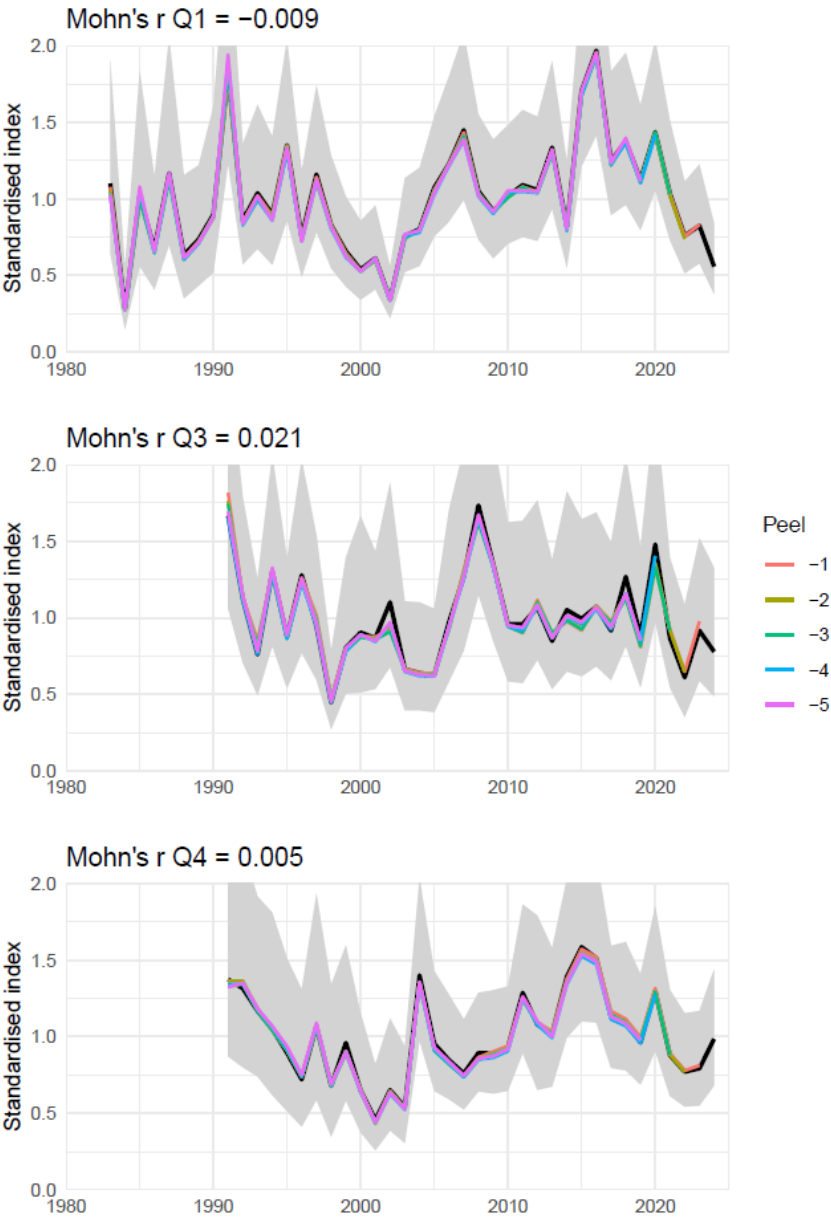


Figure 19.4. Turbot in Division 3.a. Exploitable biomass survey indices (black, base run) and 5 retrospective fits. The shaded area shows 95% confidence intervals of the base run. Top panel is quarter 1, middle panel is quarter 3 and bottom panel is quarter 4.

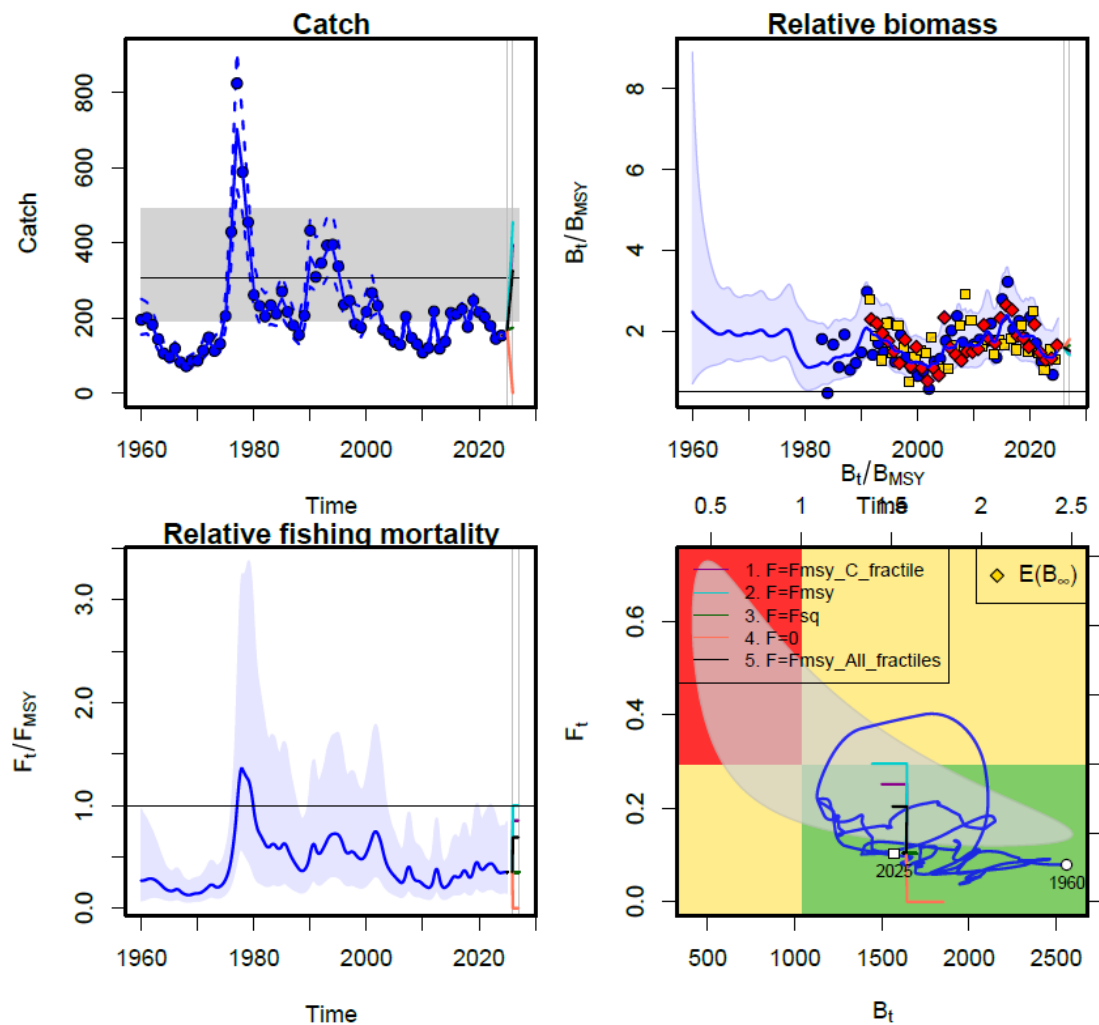


Figure 19.5. Turbot in Division 3.a. SPiCT assessment running to the end of 2024, with 5 different short term forecast scenarios. The vertical grey lines in the catch, relative biomass and relative fishing mortality plots indicate the intermediate year (2025) and the horizontal lines show the corresponding reference points (MSY , $B/B_{MSY}=0.5$ and $F/F_{MSY}=1$). The shaded areas and dashed lines in all plots show 95% confidence intervals. The assessment is based on settings agreed upon during WGNSSK 2025 (external review).

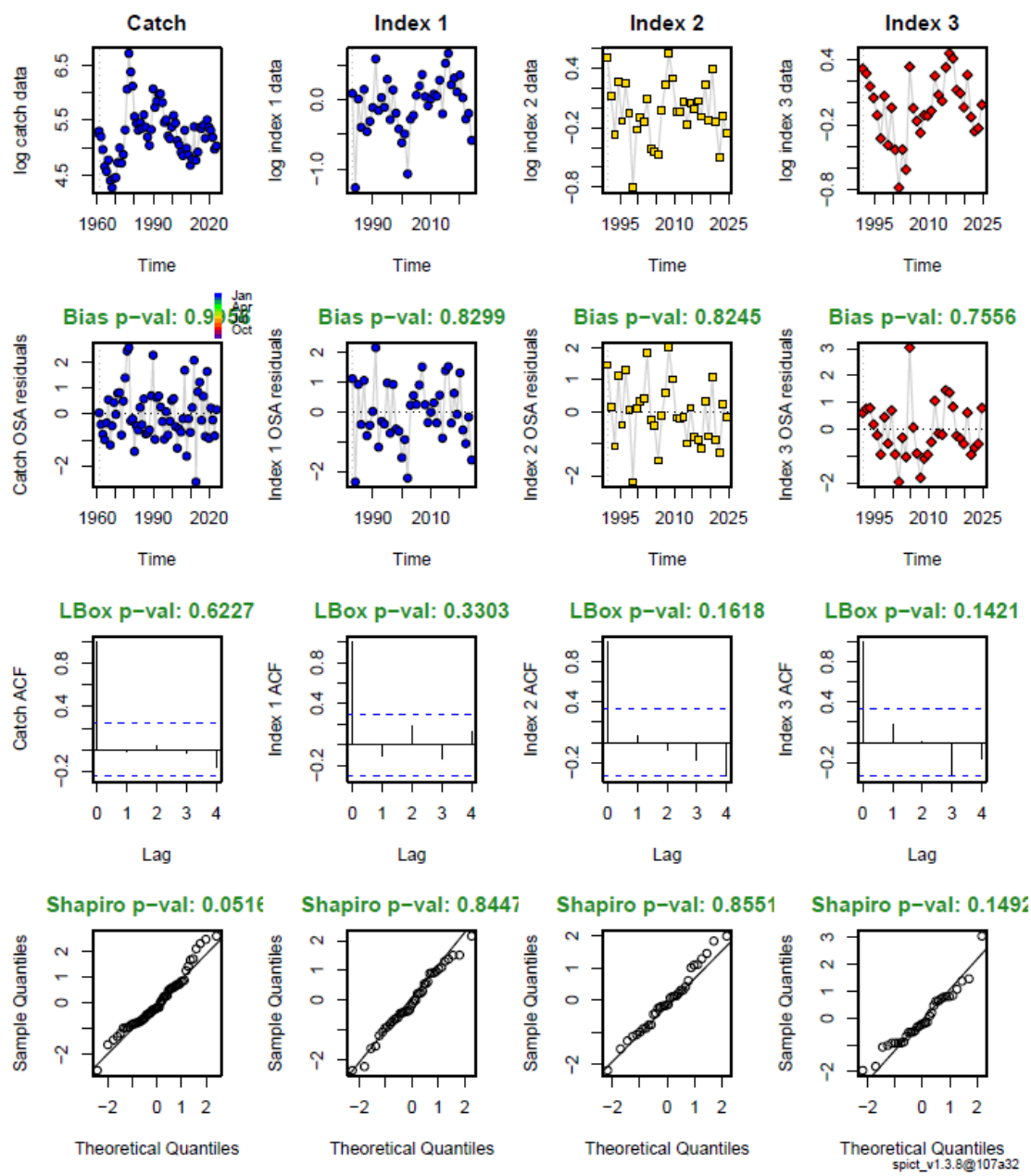


Figure 19.6. Turbot in Division 3.a. Evaluation of SPiCT assessment. The residual diagnostics are shown for the four input time series (catch: left, exploitable biomass index: other columns). From the top to bottom it is shown: the log-transformed input time series, the one-step-ahead residuals with a bias test, the autocorrelation function with a Ljung-Box test, and a QQ-plot with a Shapiro test for normality.

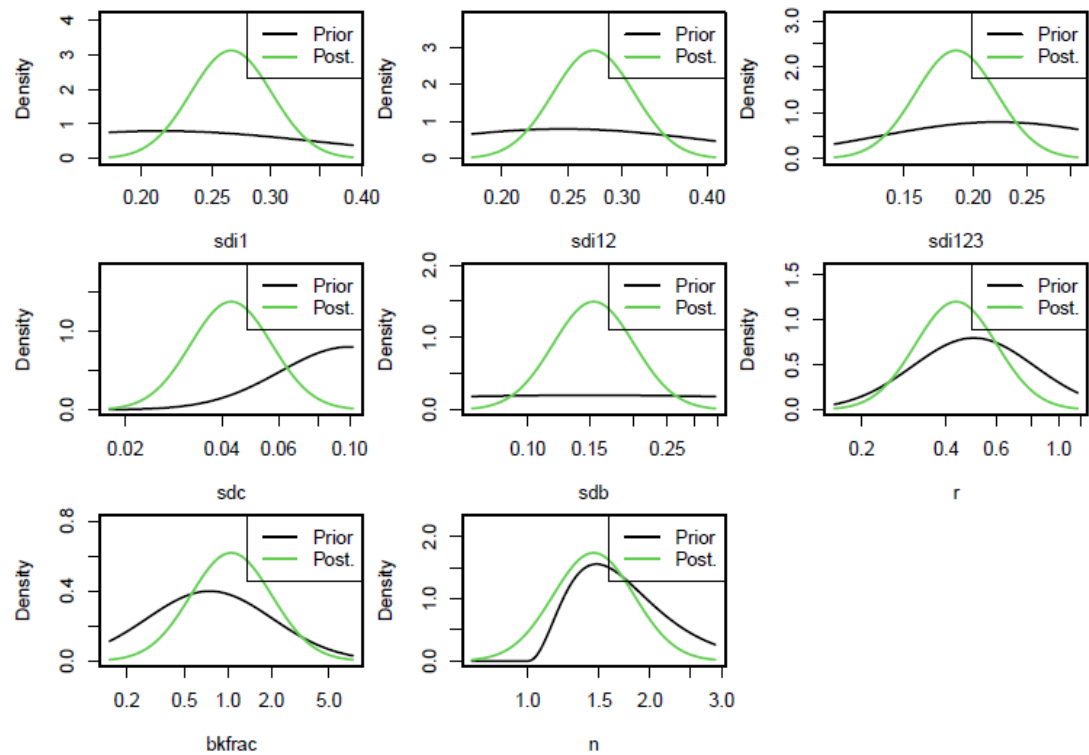
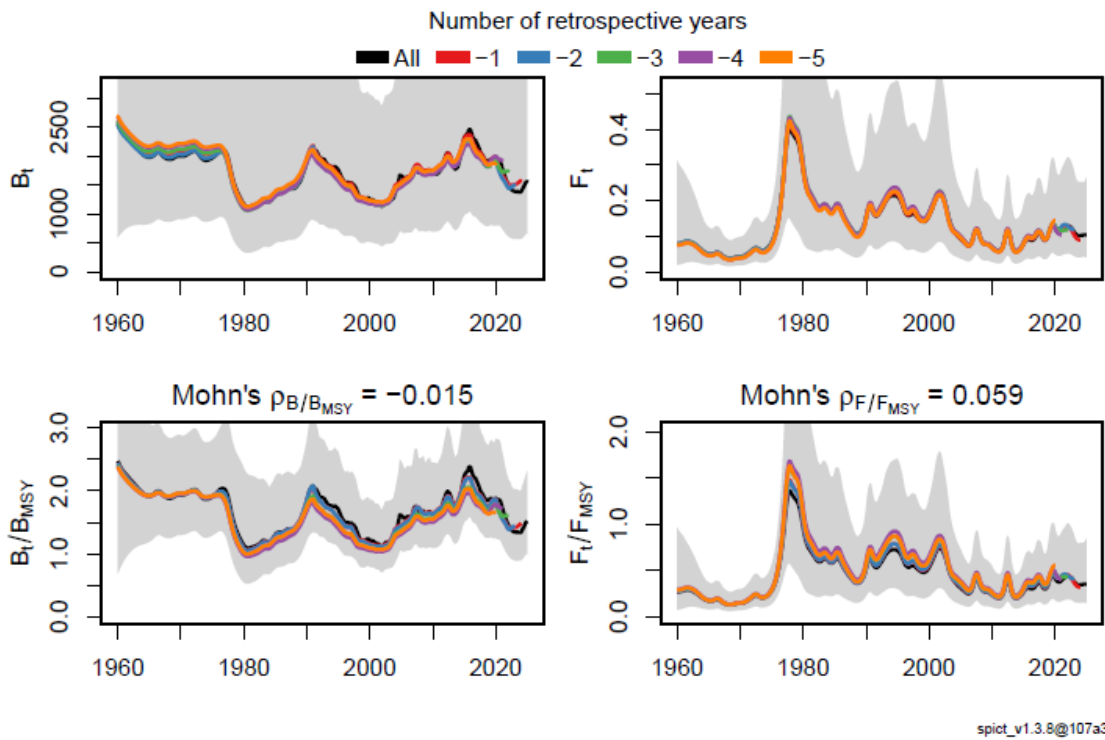


Figure 19.7. Turbot in Division 3.a. Prior (black) and posterior (green) distributions of the estimated parameters.



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Figure 19.8. Turbot in Division 3.a. Retrospective analysis showing the baseline (black lines) with 95% confidence intervals (shaded area). The survey index is re-calculated by consecutively removing the last five years of data and the assessment is shown in red, blue, green, purple and orange. The Mohn's rho for the relative quantities is shown on top of their corresponding panels.

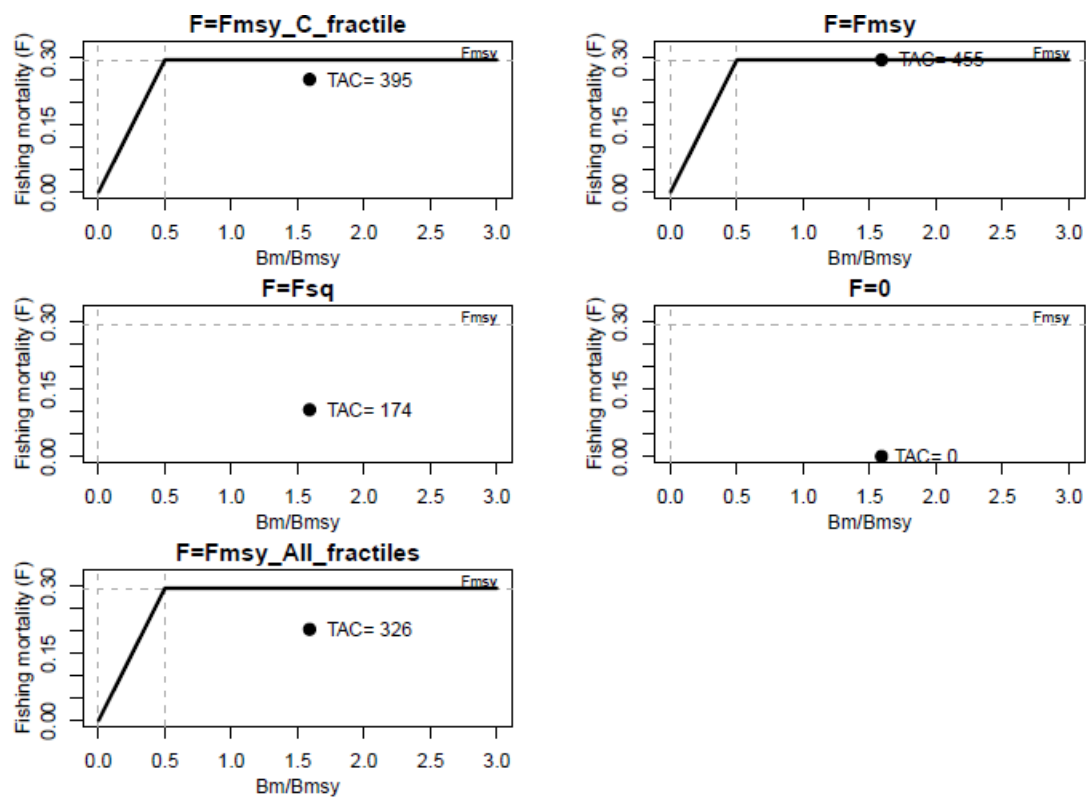


Figure 19.9. Turbot in Division 3.a. Management scenarios. The solid line shows the harvest control rule without any fractiles, i.e. fishing at F_{MSY} when B is above $B_{trigger}$, and reducing F linearly to zero when the biomass is lower. Scenarios that are based on a specific fishing mortality ($F = F_{sq}$ and $F = 0$) do not have an HCR. The vertical lines show $B/B_{trigger} = 0.5$. The basis for the advice follows that recommendation of WKLIFE X (ICES, 2020b) and is shown in the top left corner.

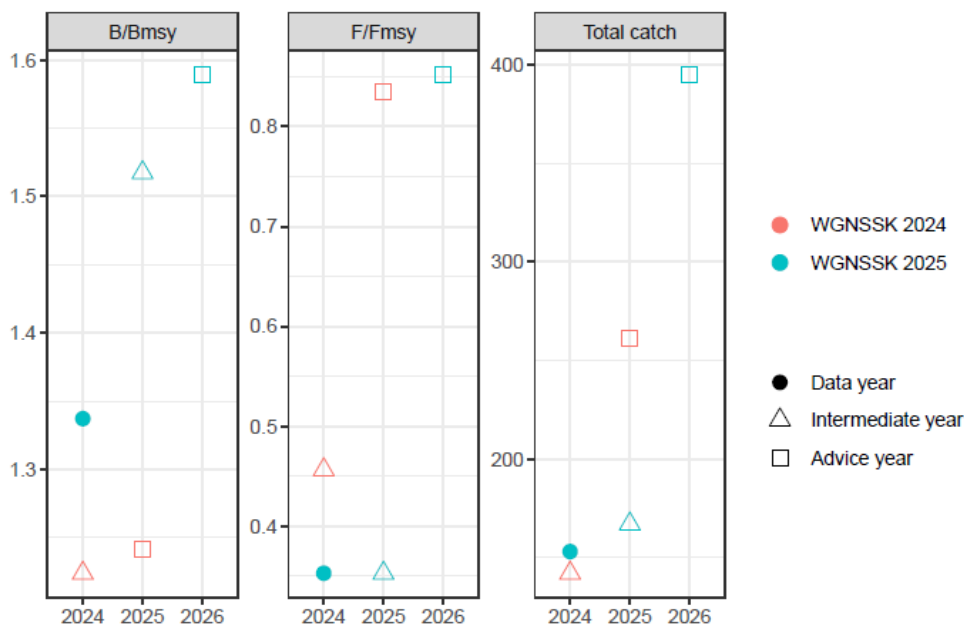


Figure 19.10. Turbot in Division 3.a. Change in advice - Comparison between last year's and this year's assessment. Note these assessment don't use the same model settings and input data (WD 8 Annex 8).

Contents

- 19 Turbot in Skagerrak and Kattegat..... 700
 - 19.1 Management regulations..... 700
 - 19.2 Fisheries data 700
 - 19.3 Survey data, recruit series and analysis of stock trends 701
 - 19.4 Assessment and short-term forecast..... 701
 - 19.5 Issue list 703
 - 19.6 Summary 703
 - 19.7 References 704
 - 19.8 Tables and figures 705